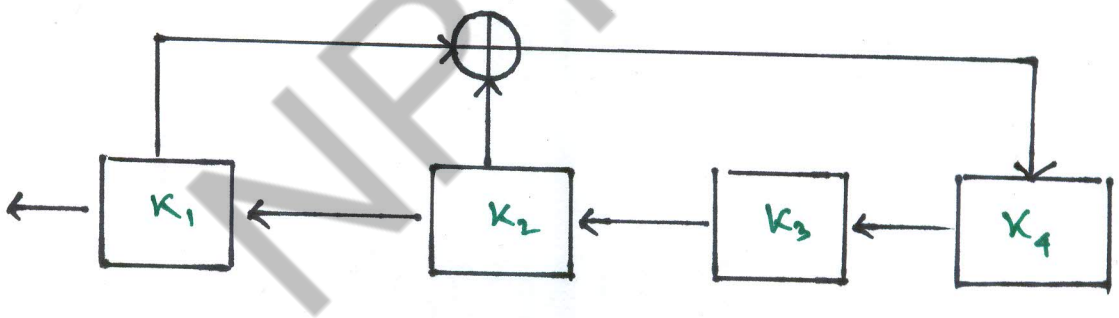
Week-3 : Lecture Notes

Topics :

LFSR based Stream cipher.
Mathematical background.
Abstract algebra.
Number theory.
Moduler inverse.

Linear Feedback Shift Register (LFSR)



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- An LFSR of length m consists of m stages numbered 1, 2, 3, ···, m, each storing one bit and having one input and one output; together with a clock which controls the movement of data.

- The vector $(K_1, K_2, \cdots, K_m)$ would be used to initialize the shift register.
- During each unit of time the following operation would be performed concurrently

  i) K_1 would be tapped as the next key stream bit.

  ii) $K_2, \cdots, K_m$ would be shifted one stage to the left

  iii) the 'new' value of K_m would be computed to be

$$\sum_{j=1}^{m-1} c_j K_{j+1}$$

The linear feedback is carried out by taping certain stages of the register (as specified by the constants c_j having the value 1') and computing a sum modulo 2 (which is an exclusive-or).

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Mathematical background

- In order to understand some of cryptographic algorithms dealt with throughout this course, it is necessary to have some background in two areas of mathematics

  1. Number Theory,
  2. Abstract Algebra.

- New Advanced Encryption standard (AES) relies on the subject of finite fields which forms a part of abstract algebra.
- Going to deal with Number Theory for the moment.

Number Theory

- Number theory deals with the theory of numbers and is probably one of the oldest branches of mathematics.

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- It is divided into several areas including elementary, analytic and algebric number theory.

- These are distinguished more by the methods used in each than the type of problems posed.

- Relevant ideas discussed here and include:
  - prime numbers
  - the greatest common divisor
  - the modulus operator
  - the moduler inverse

Prime numbers

- A prime number p is an integer greater than 1 with only two positive divisors, 1 and itself.

- Therefore it's entire set of divisors (i.e it's factor) consist only of four integers ± 1 and $\pm p$

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- It can be seen that 1 is not a prime number.
- Prime numbers are of the utmost importance to certain cryptographic algorithms and most of the techniques used will not work without them.
- Any positive integer $I \geq 2$ is either a prime or can be expressed as the product of primes.
- This is known as the fundamental theorem of arithmetic:

$$I = P_N^{\epsilon_N} \times P_{N-1}^{\epsilon_{N-1}} \times \cdots \times P_1^{\epsilon_1}, \quad P_N > P_{N-1} > \cdots > P_1 \quad \text{---- (i)}$$

- Another way of looking at this would be:

$$I = \prod_{s} P_n^{\epsilon_n} \qquad \epsilon_n \geq 0 \quad \text{---- (ii)}$$

Here 's' is the set of all prime numbers.

- In general most of the exponents ϵ_n will be '0'

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- As a result of equation (i) and (ii) any integer > 1, that is not a prime number is known as a composite number.

- It can be seen that from this and the definition of a prime number above, that '1' is neither prime nor composite.

- The first prime numbers are:
  2, 3, 5, 7, 11, 13, 17, 19, 23 and 29.

Division

- Any integer can be expressed as $n = q \times m + r$, where n, q, and r are integers, m is a positive integer and $0 \leq r < m$

- The remainder (also known as residue) r, must be non-negative (i.e either positive or 0)

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- This is seen by two restrictions:
  1. $0 \leq r < m$
  2. $q = \lfloor \frac{n}{m} \rfloor$

- The notation $\lfloor x \rfloor$ is known as the floor of the integer x and is the greatest integer $\leq x$.

- The notation $\lceil x \rceil$ is the ceiling of the integer x and is the least integer $\geq x$

- For example, $24 \div 10$ is 2 with a remainder 4 however, $-24 \div 10$ is -3 with a remainder 6 and not -2 with a remainder -4 as might be expected.

- If $r = 0$ then n is said to be a multiple of m. This is also the same as saying that m divides n, is a divisor of n or is a factor of n and the notation used to express this is $m|n$.

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- The greatest common divisor, m_{max}, of two integers a and b is the largest positive integer that will divide both a and b without a remainder.

- Therefore $m_{max} | a$, $m_{max} | b$ and $m_n | m_{max}$ for any divisor m_n of a and b.

- The notation generally used to represent this is $gcd(a, b) = m_{max}$.

- If $gcd(a, b) = 1$, this means that a and b have no common factors other than 1.

- Such pair is known as relatively prime or co-prime

- Along with prime numbers, numbers that are relatively prime have considerable importance in cryptography as will be seen later.

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- The greatest common divisor of two positive integers a and b ($\gcd(a,b)$) can be determined by a procedure known as Euclid's Algorithm. It is based on the theorem that

$$\gcd(a,b) = \gcd(b, a \bmod b)$$

- The expression 'mod' is used in moduler arithmetic which is a special kind of arithmetic involving remainders as will be seen next.

Moduler Arithmetic

- The symbol used ($\equiv$) is known as the congruence symbol.

- Moduler relationships are of the form $n \equiv R \pmod m$ (spoken as "n is congruent to R mod m") where n and R are integers and m is a positive integer known as the modulus.

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- If this congruence relationship holds, then it is said that n is congruent to R modulo m.

- The modulus operator (mod) produces the remainder when the integer on it's left is divided by the modulus.

- Thus the term (R mod m) is equal to the remainder r, when R is divided by m.

- If two remainders are equal then it can be written that $(n \bmod m) \equiv (R \bmod m)$ - a standard equality.

- However, if the modulus is equal on both sides of the equation, then the (mod m) term can be removed from the left side and the equality symbol replaced with a congruence symbol (along with a slight re-arrangement of the brackets).

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- Assuming $n \neq r$, it would be incorrect to say $n = (R \bmod m)$.
- However, it is correct to say that $n \equiv R \pmod m$ and this basically states that the same remainder (in this case r) results when both n and R are divided by m.
- As mentioned, the remainder n is also known as a residue.
- If $R = r$ (i.e $0 \leq R < m$) then R is known as a least residue
- The congruent relation '$\equiv$' is an equivalence relation.
- The set of numbers congruent to some value 'a' $\pmod m$ is known as a residue class (or a congruent class)
- As $0 \leq r < m$, this means there are m possible values of r and hence there are m possible residue classes.

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- The congruence relationship $n \equiv R \pmod{m}$ is only true if $m \mid n-R$

- If $a \equiv a_1 \pmod{m}$ and $b \equiv b_1 \pmod{m}$, then $a+b \equiv a_1+b_1 \pmod{m}$ and $ab \equiv a_1 b_1 \pmod{m}$

- The integers modulo m, denoted Z_m is the set of integers $\{0, 1, 2, \ldots, m-1\}$

- To understand why, it must be remembered that the integers n and R can be expressed as $q_{\{n,R\}} \times m + r_{\{n,R\}}$ where the subscript $\{n,R\}$ represents the fact that q and r will generally take on different values for n and R.

- Only if $r_n = r_R$ will $m \mid (n-R)$ because in this case the two remainders cancel each other in the $(n-R)$ term:

$$(q_n \times m + r_n - q_R \times m - r_R) = (q_n \times m - q_R \times m)$$

- Because $m \mid (q_n \times m)$ and $m \mid (q_R \times m)$ $\Rightarrow m \mid (q_n \times m - q_R \times m)$. If $(n-R)$ is not

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divisible by m then the notation used to represent this is $\not|$ and therefore $m \not| (n-R)$. In this case $n \not\equiv R \pmod{m}$.

Moduler Inverse

- The idea of an inverse is important both in ordinary arithmetic and moduler arithmetic.

- In any set of numbers, the inverse of a number contained in that set is another number which when combined with the first under a particular operation will give the Identity element for that operation.

- The identity element for a particular operation is a number that will leave the original number unchanged under that operation.

- Two example of inverses are the: (1) Additive inverse, (2) Multiplicative inverse.

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- The Identity element under different operations will be different.
- Under addition, it is '0', as any number added to '0' will remain unchanged.
- However, under normal multiplication the identity element is 1 as any number multiplied by 1 will remain unchanged.
- In ordinary arithmetic if the number is x then the additive inverse is $-x$ and multiplicative inverse is $\frac{1}{x}$.
- The idea is same in modular arithmetic however if x is an integer then its multiplicative inverse would not be $\frac{1}{x}$ as there is no such things as a fraction in modular arithmetic.
- In this case, it would be a number which, when multiplied by the

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original number, would give a result that is congruent 1 (mod m) (again, m is the modulus).

- A number x can only have a multiplicative inverse if it is relatively prime to the modulus (i.e $\gcd(x, m) = 1$)

- When one number is operated on modulo some other number, it is said that the first number has been reduced modulo the second and the operation is called a modular reduction.

- Let $a \in Z_m$, the multiplicative inverse of a modulo m is an integer x belongs to Z_m such that $ax \equiv 1 \pmod{m}$. If such an 'x' exists, then it is unique and a is said to be invertible. The inverse of a is denoted by a^{-1}.

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Abstract Algebra

- Will only be looking at a very small subset of what this subject has to offer.

- Three main ideas here that need to be grasped:
  1. Group $(G, \cdot)$
  2. Ring $(Rg, +, \times)$
  3. Field $(F, +, \times)$

- Basically, three different types of sets along with some operations.

- The classification of each set is determined by the axioms which is satisfies.

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Group

- A Group $\{G, \cdot\}$ is a set under some operation $(\cdot)$ if it satisfies the following 4 axioms:
  1. closure (A_1): For any two elements $a, b \in G$, $c = a \cdot b \in G$.

2. Associativity (A_2) : For any three elements $a, b, c \in G$,

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

3. Identity (A_3) : There exists an Identity element $e \in G$ such that $\forall a \in G$, $a \cdot e = e \cdot a = a$.

4. Inverse (A_4) : Each element a in G has an inverse i.e $\forall a \in G$ $\exists \bar{a}^1 \in G$, $a \cdot \bar{a}^{-1} = \bar{a}^{-1} \cdot a = e$

- However it is said to be an Abelian group if in addition to the above the set follows the axiom :

5. Commutativity (A_5) : For any $a, b \in G$

$$a \cdot b = b \cdot a$$

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Cyclic group

- Exponentiation is the repeated application of the group operator.

- We might have a^3 and this would equal $a.a.a$.
- So, if the operation was addition then a^3 would infact be $a+a+a$.
- Also we have $a^0 = e$ which for an additive group is 0.
- $a^{-n} = (a^{-1})^n$
- A group is said to be cyclic if every element of the group G is a power a^k (where k is an integer) of a fixed element $a \in G$.
- The element a is said to generate G or be a generator of G.
- A cyclic group is always abelian and may be finite or infinite.
- If a group has a finite number of elements it is referred to as a finite group.
- The order of the group is equal to the number of elements in the

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group. Otherwise the group is an infinite group.

Ring

- A binary operation is a mapping of two elements into one element under some operation. For a set S we have $f : S \times S \longrightarrow S$

- A ring $\{Rg, +, \times\}$ is a set with two binary operations addition and multiplication that satisfies the following axioms:

1. Abelian group under addition ($A_1 \rightarrow A_5$)
   It satisfies all ~~the~~ of the axioms for an abelian group (all of the above) with the operation of addition.
   The identity element is '0' and inverse is denoted by $-a$.

2. Closurer under multiplication (M_1):
   For any $a, b \in Rg$,
   $c = a \cdot b \in Rg$

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3. Associativity of multiplication (M_2):
For any elements $a, b, c \in Rg$, $(a \cdot b) \cdot c = a(bc)$

4. Distributive (M_3): For any elements $a, b, c \in Rg$, $a(b+c) = ab + ac$

- It is then said to be a commutative ring if in addition the ring follows the axiom:

5. Commutativity (M_4): For any $a, b \in Rg$ $ab = ba$

- It is an integral domain if in addition the commutative ring follows the axioms:

6. Multiplicative Identity (M_5): There is an element 1 in Rg such that $a1 = 1a = a \quad \forall \; a \in Rg$.

7. No Zero Divisors (M_6): If $a, b \in Rg$ and $ab = 0$ then either $a = 0$ or $b = 0$.

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Field

- A field $\{F, +, \times\}$ is a set with two binary operations addition and multiplication that satisfies the following axioms:

  1. Integral Domain ($A_1 \rightarrow M_6$): It satisfied all of the axioms for an integral domain (all of the above)

  2. Multiplicative Inverse (M_7): Each element in F (except 0) has an inverse i.e
     $\forall\, a(\neq 0) \in F, \exists\, \bar{a}^{-1} \in F,\ a\bar{a}^{-1} = \bar{a}^{-1}a = 1$.

- In ordinary arithmetic it is possible to multiply both sides of an equation by the same value and still have the equality intact.

- Not necessarily true in finite arithmetic.

- In this particular type of arithmetic we are dealing with a set containing a finite number of values.

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- The set of real numbers is an infinite set and is not really useful for working with an computer systems due to the limited amount of memory and processing power.

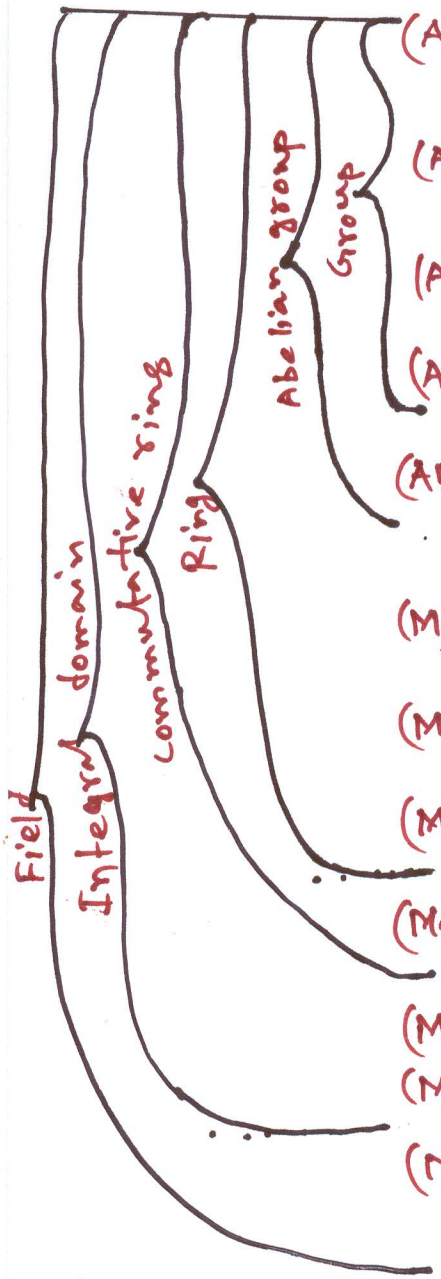
- Much easier if every operation the computer performed resulted in a finite value that was easily handled. This is where finite fields come into play.

- Closure is the property that causes the result of a binary operation on an ordered pair of a set to be a part of that set also.

- The term ordered pair is important as it is not generally the case that $a \cdot b = b \cdot a$

- To restate, Groups, Rings and Fields are all sets defined with either one or more binary operations.

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(A1) closure under addition — if $a, b \in S$ then $a+b \in S$

(A2) Associativity of addition — $(a+b)+c = a+(b+c) \quad \forall a, b, c \in S$

(A3) Additive identity — $\exists\, 0 \in S$ s.t $a+0 = 0+a = a \quad \forall a \in S$

(A4) Additive inverse — $\forall a \in S, \exists\, -a \in S$ s.t $a+(-a) = (-a)+(a) = 0$

(A5) commutativity of addition — $a+b = b+a \quad \forall a, b \in S$

(M1) closure under multiplication — if $a, b \in S$ then $ab \in S$

(M2) Associativity of multiplication — $a(bc) = (ab)c \quad \forall a, b, c \in S$

(M3) Distributive law — $a(b+c) = ab+ac \quad \forall a, b, c \in S$
$(a+b)c = ac+bc \quad \forall a, b, c \in S$

(M4) commutativity of multiplication — $ab = ba \quad \forall a, b \in S$

(M5) Multiplicative identity — $\exists\, 1 \in S$ s.t $1a = a1 = a \quad \forall a \in S$

(M6) No zero divisor — if $a, b \in S$ and $ab = 0$ then either $a = 0$ or $b = 0$.

(M7) Multiplicative inverse — $a (\neq 0) \in S, \exists\, a^{-1} \in S$ s.t. $a \cdot a^{-1} = a^{-1} \cdot a = 1$

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Fig2 Group, Ring, Field.

- Fig2 summarises the hierarchical structure of the group, ring and field. It can be seen that the group is defined under addition.